Requirement Analysis

QADemo E-Commerce Application

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Introduction

The purpose of this Requirement Analysis document is to analyze, clarify, structure and refine the business requirements of the QADemo E-Commerce Application into detailed and testable system requirements.

The analysis identifies the functional requirements, non-functional requirements, business rules, user roles, dependencies, assumptions, constraints and potential requirement ambiguities that should be considered during development and software quality assurance activities.

This document will serve as a bridge between the Business Requirements Document (BRD) and subsequent SQA activities such as test scenario design, test case development, test execution, defect reporting and requirement traceability.

Product Context

QADemo is an e-commerce application designed to simulate a real-world online shopping experience. Customers can browse products, search for products, view product information, manage shopping carts, proceed through checkout, place orders and view order-related information.

The application also provides administrative capabilities for authorized users to manage relevant products and orders.

The primary customer journey is:

Login → Browse/Search → Product Details → Cart → Checkout → Payment → Order Placement → Confirmation

This journey is consistent with the Product Overview and represents the primary business flow of the application.

Requirement Analysis Objectives

The requirement analysis aims to:

- Identify all major functional requirements of the application.
- Convert high-level business requirements into specific system requirements.
- Identify functional and non-functional expectations.
- Identify business rules that affect system behavior.
- Identify input, processing and output requirements.
- Identify positive and negative scenarios.
- Identify dependencies and assumptions.
- Identify requirement ambiguities that require clarification.
- Establish a clear basis for test scenario and test case development.
- Support traceability between business requirements and QA activities.

Stakeholder and User Analysis

Stakeholder/User	Role in the System	Main Requirements
Customer	Purchases products	Registration, login, product discovery, cart, checkout, payment, order management
Administrator	Manages application operations	Product and order management
Business Owner	Defines business objectives	Reliable purchasing and business processes
Product Manager	Defines product needs and priorities	Product discovery and customer experience
Business Analyst	Analyzes requirements	Requirement clarification and validation
Development Team	Implements the system	Builds functionality according to approved requirements
QA/SQA Team	Validates the system	Ensures requirements are correctly implemented
UI/UX Team	Designs user experience	Ensures usability and understandable interfaces

User Roles

Customer

A customer can:

- Log in and log out.
- Manage account information.
- Browse available products.
- Search for products.
- View product details.
- Check product availability.
- Add products to the shopping cart.
- Update product quantities.
- Remove products from the cart.
- Review cart information.
- Proceed to checkout.
- Provide required delivery information.
- Select an available payment method.
- Review order information.
- Place an order.
- Receive order confirmation.

Administrator

An administrator can perform authorized administrative activities, including:

- Accessing administrative functionality.
- Managing relevant products.
- Updating product information.
- Verifying product information.
- Managing customer orders.
- Updating order status where authorized.

Administrative functionality must only be accessible to authorized users.

Functional Requirement Analysis

Functional requirements describe what the system must do.

1. Customer Authentication
Requirement ID: FR-001
Related Business Requirement: BR-001
The system shall allow registered customers and authorized administrators to securely log in and access specific functionality.

Expected Behavior
- The customer and administrator shall provide valid login credentials.
- The system shall validate the credentials.
- Successful credentials shall grant appropriate access.
- Invalid credentials shall not grant access.
- The system shall display an appropriate error message for unsuccessful authentication.
- The customer and administrator shall be able to log out.
- After logout, protected functionality shall not remain accessible without appropriate authentication.

Security Considerations
- Unauthorized users must not access protected information.
- Authentication errors should not expose sensitive account information.
- Session behavior should be handled appropriately.

2. Product Discovery
Requirement ID: FR-002
Related Business Requirement: BR-002
The system shall allow customers and administrators to discover available products through browsing.

Expected Behavior
The customer and administrator shall be able to:
- View available products.
- Browse products by category where categories are supported.
- Navigate between product listings and product details.

3. Product Search
Requirement ID: FR-003
Related Business Requirement: BR-003
The system shall allow customers and administrators to search available products through searching.

Expected Behavior

The customer and administrator shall be able to:

- Search for available products.
- View relevant search results.
- Open a product from the product listing.

Search Testability Considerations

The system should appropriately handle:

- Exact match.
- Partial match.
- Case sensitivity.
- No matching product.
- Empty search.
- Special or invalid search.

4. Product Information

Requirement ID: FR-004

Related Business Requirement: BR-004

The system shall provide relevant product information before the customer makes a purchasing decision.

Product Information May Include

- Product name.
- Product image.
- Product description.
- Product price.
- Product category.
- Product availability/stock information.
- Other relevant product attributes.

Expected Behavior

The displayed product information should be:

- Correct.
- Consistent.
- Readable.
- Relevant to the selected product.
- Consistent between product details and checkout/cart information.

5. Shopping Cart

Requirement ID: FR-005

Related Business Requirement: BR-005

The system shall allow customers to add and manage selected products before checkout.

Expected Behavior

The customer shall be able to:

- Add a product to the cart.
- Add multiple products.
- View cart contents.
- Update product quantity.
- Remove a product from cart.
- Review product prices.
- Review quantities.
- Review the cart total.
- Continue shopping.
- Proceed to checkout.

Cart Testability Considerations

The system should correctly handle:

- Quantity = 1.
- Quantity greater than 1.
- Invalid quantity.
- Zero quantity.
- Removing the only item.
- Empty cart.
- Multiple products.
- Same product added multiple times.
- Unavailable/out-of-stock product.

6. Order Calculation

Requirement ID: FR-006

Related Business Requirement: BR-006

The system shall accurately calculate the total value of products selected by the customer.

Calculation Requirements

The system shall calculate:

Item Total = Product Price × Quantity

Order Total = Sum of all applicable item totals + applicable charges − applicable discounts

Where applicable, the system should also correctly calculate:

- Subtotal.
- Delivery/shipping charge.
- Discount.
- Tax.
- Final payable amount.

If a particular component is not supported by the application, it should not be assumed to exist.

Validation

The total displayed in:

Product → Cart → Checkout → Order Confirmation

should remain consistent unless a valid business rule changes the amount.

7. Checkout

Requirement ID: FR-007

Related Business Requirement: BR-007

The system shall allow customers to proceed from the shopping cart to checkout and provide the information required to complete an order.

Expected Behavior

The customer shall be able to:

- Proceed from cart to checkout.
- Provide required customer information.
- Provide required delivery information.
- Select an available payment method.
- Review selected products.
- Review quantities.
- Review prices.
- Review total amount.
- Correct invalid information before confirmation.
- Proceed to final order confirmation.

Validation

The system shall validate:

- Required fields.
- Invalid input.
- Missing delivery information.
- Invalid payment information where applicable.
- Invalid order information.

8. Order Placement

Requirement ID: FR-008

Related Business Requirement: BR-008

The system shall allow customers to place an order for eligible products after completing the required checkout process.

Preconditions

Before an order can be placed:

- The customer must have the required access.
- The cart must contain an eligible product.
- Required checkout information must be provided.
- Required payment processing must be completed or accepted.
- The order information must be valid.

Expected Result

After successful order placement:

- The order should be created.
- The customer should receive appropriate confirmation.
- Relevant order information should be available.
- The order should enter the appropriate order status.

9. Payment

Requirement ID: FR-009

Related Business Requirement: BR-009

The system shall support the payment process required to complete an order.

Expected Scenarios

The system should support appropriate handling of:

- Successful payment.
- Failed payment.
- Invalid payment information.
- Cancelled payment.
- Payment timeout, where applicable.
- Payment service failure, where applicable.

Business Impact

Payment failures may result in lost sales and customer dissatisfaction; therefore, successful and unsuccessful payment scenarios must be validated thoroughly.

10. Order Confirmation

Requirement ID: FR-010

Related Business Requirement: BR-010

The system shall provide confirmation after a successful order.

Expected Confirmation Information

Where supported, confirmation should include:

- Order ID/reference.
- Order status.
- Ordered products.
- Quantities.
- Order amount.
- Customer/delivery information.
- Payment status.
- Confirmation message.

The system must not indicate that an order was successfully placed when the required order process has not actually completed.

11. Product Administration

Requirement ID: FR-011

Related Business Requirement: BR-011

The system shall allow authorized administrators to manage relevant product information.

Administrator Activities

Where supported, administrators should be able to:

- Add products.
- Update products.
- Verify product information.
- Manage product availability.
- Maintain accurate product information.

Unauthorized users must not be able to perform administrative product-management activities.

12. Order Management

Requirement ID: FR-012

Related Business Requirement: BR-012

The system shall allow authorized administrators to manage orders.

Administrator Activities

Where supported, administrators should be able to:

- Update orders.
- Verify order information.
- Manage order status.

Unauthorized users must not be able to perform administrative order-management activities.

13. Input Validation Requirements

Requirement ID: FR-013

Related Business Requirement: BR-001, BR-003, BR-005, BR-007, BR-009, BR-011, BR-012

The application shall validate user-provided information before processing it.

Validation Areas

Validation should be applied to:

- Login fields.
- Search fields.
- Quantity fields.
- Delivery information.
- Payment information.
- Administrative product information.
- Other applicable input fields.

General Validation Rules

The system should:

- Identify missing required information.
- Reject invalid formats.
- Reject invalid values.
- Display understandable validation messages.
- Prevent invalid data from being processed.

Business Rule Analysis

The following business rules have been identified from the Business Requirements Document.

Rule ID	Business Rule	Requirement Impact
BRULE-001	Customers can only order eligible products.	Cart and Order Placement
BRULE-002	Checkout prices must accurately reflect applicable product prices.	Cart, Checkout, Order
BRULE-003	An order is not successful until the required order process is completed.	Order Placement
BRULE-004	Customers receive confirmation after successful ordering.	Order Confirmation
BRULE-005	Only authorized users can access administrative functionality.	Authentication/Admin
BRULE-006	Customer and order information must be appropriately protected.	Security
BRULE-007	Information should remain accurate throughout shopping and ordering.	Product, Cart, Checkout, Order

Non-Functional Requirement Analysis

Non-functional requirements describe how well the system should operate.

1. Usability
NFR-001
The application should provide a usability expectations or evaluation criteria.
The customer should be able to understand:

- Where to find products.
- How to search.
- How to add products to the cart.
- How to modify the cart.
- How to proceed to checkout.
- How to complete an order.
- How to view order information.

Poor usability may increase customer abandonment, which is identified as a business risk in the BRD.

2. Performance

NFR-002

The application should respond within a response-time targets during normal usage.

Performance should be considered for:

- Page loading.
- Product search.
- Product details.
- Cart updates.
- Checkout.
- Order placement.

Specific performance targets should be defined by the project stakeholders if required.

3. Security

NFR-003

The system should protect customer and business information from unauthorized access.

Security should cover:

- Authentication.
- Authorization.
- Customer account information.
- Order information.
- Administrative functionality.
- Payment-related information.

4. Reliability

NFR-004

The application should perform critical business transactions.

Critical workflows include:

Login → Product Discovery → Cart → Checkout → Payment → Order Placement → Confirmation

Failure in a critical workflow should not result in an incorrect order state or misleading customer confirmation.

5. Compatibility

NFR-005

The application should operate correctly on supported browsers and devices (like Crome, Firefox, iPhone, Android, OS, Windows etc.) defined by the project.

Compatibility testing should be performed based on the project's supported environment.

6. Data Integrity

NFR-006

The application should maintain consistent and accurate information throughout the customer purchasing journey.

For example:

Product Price = Cart Price = Checkout Price = Order Price

unless an explicitly defined business rule changes the value.

Requirement Dependencies

The identified requirements have several dependencies.

Dependency	Related Requirements
Authentication/authorization	Customer/Admin functionality
Product information	Product Discovery, Product Details
Product/inventory availability	Cart, Order Placement
Cart functionality	Order Calculation, Checkout
Payment mechanism	Payment, Order Placement
Order management	Order Confirmation, Order Tracking
Email/notification service where applicable	Order Confirmation
Application infrastructure	All modules

The BRD also identifies product/inventory information, customer accounts, order management, payment mechanisms, email/notification services, infrastructure and stakeholder availability as project dependencies.

Assumptions

The following assumptions are made during requirement analysis:

1. **User Roles:** The system has two primary roles: Customer and Administrator.
2. **Authentication:** Users must log in with valid credentials to access protected functionality.
3. **Product Availability:** Customers can purchase products that are available and in stock.
4. Order Flow: A customer must complete the required checkout information before placing an order.
5. **Payment:** An order requires a successful payment step where payment is applicable.
6. **Application Behavior:** Requirements marked as “To Verify” will be confirmed against the actual QADemo application before detailed test design.

These assumptions are consistent with the assumptions documented in the BRD.

Constraints

The requirement analysis is subject to the following constraints:

- The application must operate within the capabilities of the existing QADemo environment.
- Testing scope may be limited by available functionality.
- External services may have availability or integration limitations.
- Development and testing time may be limited.
- Available resources may limit testing depth.
- Requirements may change based on stakeholder feedback.

These constraints are identified in the existing BRD.

Requirement Ambiguity Analysis

Requirement analysis should identify statements that cannot yet be tested precisely.

Area	**Current Requirement**	**Ambiguity / Clarification Needed**
User name	Customer can login with their username	What are the login rules?
Password	Customer authentication is secure	What are the password rules?
Search	Customer can search products	Is search exact, partial, case-insensitive, or all three?
Product Availability	Customer can view availability	What statuses should be displayed?
Quantity	Customer can update quantity	What is the minimum and maximum quantity?
Pricing	Total should be accurate	Are tax, discount and shipping included?
Payment	Application supports payment	Which payment methods are supported?
Payment Failure	Payment can fail	What happens to the order after failure?
Order Status	Customer can track status	Which statuses are supported?
Admin	Admin manages products/orders	Which exact actions are permitted?
Confirmation	Customer receives confirmation	Is confirmation displayed, emailed, or both?
Compatibility	Application should work on supported devices	Which browsers/devices are officially supported?

These items should be clarified before final test case design whenever the actual application behavior does not define them.

Requirement Status Definitions

Status	Meaning
Confirmed	Clearly defined by BRD/application
Ambiguous	Requirement exists but meaning isn't clear
Assumption	Temporary assumption made because information is unavailable
To Verify	Need to check actual application behavior

Application Verification Items

ID	Area	Item to Verify	Reason
AVI-001	Login	Whether customer login is available	Login behavior is not fully defined
AVI-002	Search	Exact/partial/case behavior	Search matching rules are unclear
AVI-003	Cart	Quantity limits	Minimum/maximum quantity is undefined
AVI-004	Payment	Supported payment methods	Payment mechanism is not fully defined
AVI-005	Order	Supported order statuses	Order status behavior is unclear
AVI-006	Admin	Permitted product/order actions	Admin permissions require verification
AVI-007	Confirmation	Display/email mechanism	Confirmation mechanism is unclear
AVI-008	Compatibility	Supported browsers/devices	Environment is undefined

Requirement Priority Analysis

The project uses four priority levels:

Priority	Meaning
Critical	Essential for completing the primary business transaction
High	Important for normal customer/business operations
Medium	Valuable but does not prevent the primary transaction
Low	Nice-to-have or future functionality

This priority model is defined in the project's BRD.

Proposed Functional Priority

Requirement	Priority
Customer Authentication	Critical
Product Discovery	Critical
Product Information	High
Shopping Cart	Critical
Order Calculation	Critical
Checkout	Critical
Order Placement	Critical
Payment	Critical
Order Confirmation	High
Order Management	High
Product Administration	High
Input Validation	Critical

Requirement Analysis Matrix

BR ID	Requirement	Functional /NFR	Testability	Ambiguity	Dependency	Risk	Priority	Status
BR-001	Customer Authentication	Functional	High	Low	Account Data	High	Critical	Confirmed
BR-002	Product Discovery	Functional	High	Low	Product Data	High	Critical	Confirmed
BR-003	Product Search	Functional	Medium	Medium	Product Data	Medium	Critical	Ambiguous
BR-004	Product Information	Functional	High	Medium	Product Data	High	High	Ambiguous
BR-005	Shopping Cart	Functional	High	Medium	Product / Cart Data	High	Critical	Ambiguous
BR-006	Order Calculation	Functional	High	Medium	Cart / Pricing	Critical	Critical	Ambiguous
BR-007	Checkout	Functional	Medium	High	Cart / Delivery Data	Critical	Critical	Ambiguous
BR-008	Order Placement	Functional	High	Medium	Checkout / Order Data	Critical	Critical	Ambiguous
BR-009	Payment	Functional	Medium	High	Payment Mechanism	Critical	Critical	Ambiguous
BR-010	Order Confirmation	Functional	High	Medium	Order Data	High	High	Ambiguous
BR-011	Product Administration	Functional	Medium	Medium	Product Data / Admin Permissions	High	High	To Verify
BR-012	Order Administration	Functional	Medium	Medium	Order Data / Admin Permissions	High	High	To Verify

Positive and Negative Requirement Analysis

A professional QA analysis should consider both expected and unexpected behavior.

Positive Scenarios

Examples:

- Valid login.
- Successful product search.
- Viewing an available product.
- Adding an available product to cart.
- Updating quantity with a valid value.
- Successful checkout.
- Successful payment.
- Successful order placement.
- Successful order confirmation.
- Authorized administrator updating a product.

Negative Scenarios

Examples:

- Login with invalid credentials.
- Searching for a non-existing product.
- Adding an unavailable product.
- Invalid cart quantity.
- Checkout with missing required information.
- Invalid payment information.
- Failed payment.
- Attempting to place an order with an invalid cart.
- Unauthorized access to administrative functionality.
- Incorrect or inconsistent order information.

Requirement Traceability

Each functional requirement should be traceable back to a business requirement and forward to QA artifacts.

Business Requirement	**Functional Requirement**	**Test Artifact**
BR-001	FR-001 Authentication	Login/logout test cases
BR-002	FR-002 Product Discovery	Browse products test cases
BR-003	FR-003 Product Search	Search products test cases
BR-004	FR-004 Product Information	Product detail test cases
BR-005	FR-005 Shopping Cart	Cart test cases
BR-006	FR-006 Order Calculation	Calculation test cases
BR-007	FR-007 Checkout	Checkout test cases
BR-008	FR-008 Order Placement	Order placement test cases
BR-009	FR-009 Payment	Payment test cases
BR-010	FR-010 Order Confirmation	Order confirmation test cases
BR-011	FR-011 Product Administration	Update products info test cases
BR-012	FR-012 Order Management	Update and manage customer orders test cases
BR-001, BR-003, BR-005, BR-007, BR-009, BR-011, BR-012	FR-013 Input Validation	Input validation test cases

This traceability will allow QA to demonstrate that each business requirement has corresponding test coverage.

Critical End-to-End Business Flow

The most important flow identified from the requirements is:

- Login
- Browse/Search Product
- View Product Details
- Add Product to Cart
- Review/Update Cart
- Calculate Order Total
- Checkout
- Provide Delivery Information
- Complete Payment
- Place Order
- Receive Order Confirmation

This should receive the highest level of QA attention because failure at any critical point can prevent the primary business transaction.

Requirement-Based Risk Analysis

Requirement Area	Risk	Likelihood	Impact	Risk Score	Risk Level	QA Focus
Authentication	Unauthorized access	3	5	15	High	Authentication and authorization testing
Product Information	Incorrect product information	3	4	12	High	Data validation
Cart	Incorrect quantity/product	4	4	16	High	Cart validation
Order Calculation	Incorrect total	4	5	20	Critical	Calculation testing
Checkout	Checkout failure	3	5	15	High	End-to-end testing
Payment	Payment failure/incorrect status	4	5	20	Critical	Positive and negative payment testing
Order Placement	Incorrect/lost order	3	5	15	High	Order creation validation
Order Confirmation	False/missing confirmation	3	4	12	High	Confirmation testing
Administration	Unauthorized changes	3	5	15	High	Authorization testing
Usability	Customer abandonment	3	3	9	Medium	Usability testing

The BRD specifically identifies incorrect product information, incorrect cart/order calculations, checkout failure, payment failure, unauthorized access, order processing errors and poor usability as business risks.

Out-of-Scope Requirements

The following capabilities are outside the initial requirement scope unless specifically introduced later:

- Social media login.
- Advanced loyalty/rewards programs.
- Subscription-based purchases.
- International shipping management.
- Multi-vendor marketplace functionality.
- Advanced recommendation/personalization.
- AI-based customer support.
- Cryptocurrency payments.
- Complex business analytics and forecasting.
- Physical warehouse management.
- Supplier and procurement management.

These exclusions are defined in the project's BRD.

Requirement Validation Criteria

The analyzed requirements should be considered sufficiently validated when:

1. Registered customers can authenticate successfully.
2. Customers can browse and search available products.
3. Customers can view accurate product information.
4. Customers can add and manage products in the cart.
5. Cart calculations are accurate.
6. Customers can successfully complete checkout.
7. Valid payments can be processed according to the supported payment mechanism.
8. Eligible orders can be successfully placed.
9. Customers receive appropriate order confirmation.
10. Customers can access relevant order information.
11. Authorized administrators can perform their permitted activities.
12. Unauthorized users cannot access restricted functionality.
13. Critical business workflows operate without major business-impacting defects.
14. Product, cart, checkout, payment and order information remain consistent throughout the transaction.

QA Perspective

From an SQA perspective, the analyzed requirements provide the foundation for:

- Test scenarios.
- Test cases.
- Test data.
- Functional testing.
- Negative testing.
- Boundary value analysis.
- Equivalence partitioning.
- Validation testing.
- Integration testing.
- End-to-end testing.
- Security-related testing.
- Usability testing.
- Compatibility testing.
- Regression testing.
- Requirement Traceability Matrix (RTM).

The most important QA focus should be the *complete purchasing transaction*, because the application's core business value depends on customers being able to successfully discover a product and complete a purchase.

Requirement Analysis Summary

The analysis of the QADemo E-Commerce Application identifies the customer's complete purchasing journey as the primary functional workflow.
The major requirement areas are:

Authentication → Product Discovery → Product Information → Shopping Cart → Order Calculation → Checkout → Payment → Order Placement → Order Confirmation → Order Management

Administrative functionality provides supporting business operations through authorized product and order management.
The requirements have been analyzed from functional, non-functional, business-rule, validation, dependency, risk, ambiguity, priority and QA perspectives. This analysis provides a structured foundation for developing detailed test scenarios and test cases while maintaining traceability to the original business requirements.

Document Approval

Role	**Name**	**Approval**	**Date**
Business Owner	___________	___________	___________
Product Manager	___________	___________	___________
Business Analyst	___________	___________	___________
QA/SQA Representative	Humayra Siddika Maharun	___________	___________

End of Document